

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |       | REMARK |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------|--------|
| 9.0           | Supply, installation, testing and commissioning of Under deck insulation for exposed roof of AC area using 25 mm thick (FM approved) SS make non-fibrous, CFC free, closed cell, elastomeric nitrile rubber foamed insulation in pre-formed sheets with thermal conductivity of not more than 0.036 W/(m.k) at 100 C. The material should have high resistance to fire (melting class 0). The material used shall be in rolls with self adhesive. The insulation sheet joints shall be sealed using self adhesive black tapes.                                                                                                                                               | Sqm | 400   | Comply |
| 10.0          | <b>Supply, installation, testing and commissioning of first layer of closed cell nitrile rubber insulation on SA duct, RA duct of (FM approved) including adhesive required for insulation application, fixing of black tape as per specifications.</b>                                                                                                                                                                                                                                                                                                                                                                                                                      |     |       | Comply |
|               | Without GC cloth for Indoor application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |       | Noted  |
| 10.01         | SA duct (25mm thick)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Sqm | 25400 | Noted  |
| 10.02         | RA duct (19 mm thick)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Sqm | 25400 | Noted  |
| 11.0          | <b>With GC cloth for Outdoor application</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |       | Noted  |
| 11.01         | SA duct (25mm thick)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Sqm | 500   | Noted  |
| 11.0          | RA duct (19 mm thick)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Sqm | 400   | Noted  |
| <b>HVAC.C</b> | <b>MANIFOLDS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |       |        |
| <b>1.0</b>    | <b>Chilled Water Manifold</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |       |        |
|               | Chilled Water Manifold Piping in MS Class 'C' ERW (Conforming to IS1239 ERW) construction complete with fittings, Bends, elbows, union and support hangers, sleeves and Insulation etc.<br>Supply, Fabricate, Installation and Testing of AHU Manifold for Chilled Water comprising Isolation valves, Pressure Gauge (with 100 mm dial of 0 to 10 kg/cm <sup>2</sup> range with isolation valve & mounting accessories like nipple etc with insulation) and Temperature Gauge (Dial type Thermometer with 100 mm dial of 0-30°C range with thermowell & mounting accessories like nipple etc.) and piping insulation.<br>Manifolds to be duly supported at proper intervals. |     |       | -      |
|               | 2 Nos. Ball / Butterfly Valve type isolation valve.<br>1 No. Y-strainer with SS / Bronze mesh in supply line<br>2 Nos. Pressure Gauge (with 100 mm dial of 0 to 10 kg/cm <sup>2</sup> range with isolation valve & mounting accessories like nipple, SS syphon tube etc with insulation)<br>2 Nos. Temperature Gauge (Dial type Thermometer with 100 mm dial of 0-30°C range with thermowell & mounting accessories like nipple etc.).<br>1 No. Auto Air Vent Valve of size 25mm<br>1 No. Drain Valve of Size 25mm (to be provided at bottom of coils)<br>2 Sets of flanges to be supplied for both supply & return chilled water line.                                      |     |       | -      |
|               | Valve shall be considered with extended stem. Pipe section insulation shall be considered upto 50 mm pipe and above that shall be considered with sheet insulation.<br>The Structural Support as required to be considered as Part of Manifold and shall be grouted on the floor.<br>(Manifold upto 40 mm to be with Ball Valve and above 40 mm to be Butterfly Valve.) Control valve fitment to be part of pricings. Control valve mentioned below as a separate BOQ item.                                                                                                                                                                                                  |     |       | -      |
|               | The piping Insulation for chilled water piping shall be closed cell elastomeric nitrile rubber with factory laminated GC Cloth. Insulation have density of 40-60 Kg/m <sup>3</sup> . Thermal conductivity shall not exceed 0.035 W/m/K at mean temperature of 0°C<br>Insulation shall be FM approved.<br>Manifold insulation include valve boxes, flanges, elbows fittings up to matting flange<br>1) 19 mm thick for 15-32 NB Line Size<br>2) 25 mm thick for 40 - 250 NB Line Size                                                                                                                                                                                         |     |       | Comply |
|               | Kindly Follow CHW Piping Material Specifications (PMS) as per attached tender documents                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |       | Noted  |
| 1.01          | 20 NB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | No  | 2     |        |
| 1.02          | 25 NB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | No  | 20    |        |
| 1.03          | 32 NB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | No  | 2     |        |
| 1.04          | 40 NB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | No  | 4     |        |
| 1.05          | 50 NB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | No  | 12    |        |
| 1.06          | 65 NB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | No  | 2     |        |
| 1.07          | 80 NB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | No  | 6     |        |
| 1.08          | 100 NB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | No  | 24    |        |
| 1.09          | 125 NB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | No  | 2     |        |

